Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/12/2004
PI Name:	Banerjee, Sanjoy Ph.D.		
Project Title:	Direct Numerical Simulation of Turbulent Flows with Phase Change in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2414		
Performance Goal No.:	2.8		
Performance Goal Text:	Establish the knowledge base required to design chemical process systems for exploration missions.		
	The basic flow configuration is between two parallel plates. The plates will include a two phase mixture and a shear flow induced by a pressure gradient or by the relative motion of the parallel plates. The flow will be directly simulated with resolution to resolve the surface tension dominated scales. The direct numerical simulations (DNS) will be fully 3D and established techniques (such as projection method) will be used to solve the governing conservation equations with the appropriate wall conditions. To model interface behavior we are using the phase field models that can model phase separation and surface tension effects. Inlet and outlet conditions will be periodic as in most DNS of this type. The objective is to develop and demonstrate fully-resolved direct numerical simulations (DNS) of micro-gravity multiphase flows. Surface tension effects are expected to affect flow regimes, turbulence structure, and interphase transport processes more in space than on Earth. Many of the important surface tension effects occur at micro and		

Task Description:	nano-scale dimensions. Phase density differences, instead, affect flow regimes on Earth more than on space. These DNS would therefore serve to elucidate flow microphysics that are of importance in space but difficult to extract from Earth-based experiments and should clarify critical phenomena that require microgravity experiments for elucidation. The proposed DNS methodology, when developed, would provide NASA with a tool to complement and aid in the design of flight experiments. Multiphase flows occur in a variety of equipment involved in space thermal management and life support, e.g., in evaporators, condensers, electrochemical gas generators and space based chemical processes in general. Compared to such flows on Earth, surface tension forces play a much more significant role in space while density gradients are more important on Earth. Even at relatively high Reynolds numbers the phases are distributed differently from on Earth. In view of this, experiments in space are essential, but are expensive to do. Clearly computations of such flows require adequate models, which must primarily be based on experiment because of the complexity of the problem. Direct numerical simulations are increasingly used to improve understanding and provide databases to aid in the development of such models, and hence play a complementary role to experiments. Should the simulations be successful, NASA will have a significant tool that might reduce the number of space experiments needed, help in the design of those that are essential, and provide guidance in the development of simpler models for multiphase flows – all matters of importance in achieving its mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have set a new numerical procedure based on phase field models to simulate two phase flow. Phase fields models offer a systematic physical approach for investigating complex multiphase systems such as near critical interfacial behavior, phase separation under shear flow, and microstructure evolution during solidification. However, because interfaces are replaced by thin transition regions (diffuse interface), phase field simulations require resolving very thin layers to capture the physics of the problems studied. This demands robust numerical methods that can achieve efficiently high resolution and accuracy, especially in three dimensions. We developed an accurate and efficient numerical method to solve the coupled Cahn-Hilliard/Navier-Stokes system, known as Model H, that constitutes a phase field model for density-matched binary fluids with variable mobility and viscosity. The numerical method is a timesplit scheme that combines a novel semi-implicit discretization for the convective Cahn-Hilliard equation with state-of-the-art high resolution CFD schemes employed for direct numerical simulations of turbulence. This new semi-implicit discretization is simple but effective since it removes the stability constraint due to the nonlinearity of the Cahn-Hilliard equation at the same cost of an explicit scheme. It is derived from a discretization used for diffusive problems that we further enhance to efficiently solve variable mobility and viscosity fluid flow problems. For a channel geometry, the method uses a spectral discretization in the streamwise and spanwise directions and a combination of spectral and “spectral-like” compact finite difference discretization in the wall normal direction. The capabilities of the method are demonstrated with several examples including phase separation with and without shear in two and three dimensions. The method effectively resolves interface layers of as few as three mesh points. The numerical examples show agreement with analytical solutions and scaling laws where available and the 3D simulations in the presence of shear flow reveal rich and complex structures characterized by a combined formation of plates and strings.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Badalassi VE, Ceniceros HD, Banerjee S. "Gravitational effects on structure development in quenched complex fluids." Ann N Y Acad Sci. 2004 Nov;1027:371-82. https://doi.org/10.1196/annals.1324.031 ; PubMed PMID: 15644369 , Nov-2004